

Employee Salary & Headcount Dashboard

Data Analysis Report | Prepared by Hassan Jumaa (Riq)

Project Overview

This report summarizes an end-to-end employee data analysis project: cleaning a raw HR dataset in Python (Pandas), loading it into a MySQL database, querying it with SQL, and visualizing the results in an interactive Power BI dashboard. The dataset covers 9 employees across 5 Kenyan cities (Nairobi, Mombasa, Kisumu, Nakuru, Eldoret).

Tools & Workflow

Stage	Tool	Purpose
Data Cleaning	Python (Pandas, NumPy)	Remove duplicates, handle nulls, fix invalid ages, cap salary outliers
Storage	MySQL	Persist the cleaned dataset for querying and reporting
Querying	SQL	Aggregate and rank employees by salary, city, and hire date
Visualization	Power BI	Interactive dashboard with KPI cards, charts, and slicers

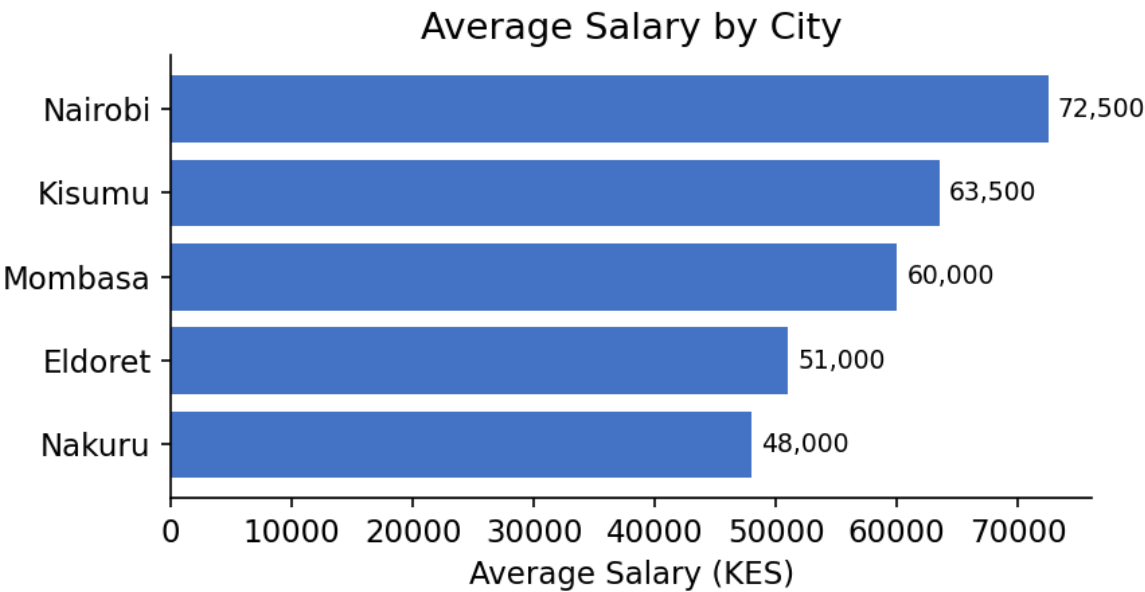
Data Cleaning Summary

Issue Found	Resolution
Duplicate employee records	Dropped exact duplicate rows
Missing employee name	Filled with placeholder, later corrected to 'Jamal'
Missing age values	Filled with median age
Invalid negative age (Faith, -5)	Reset to missing, then re-filled with median and rounded to whole years
Missing salary value	Filled with median salary
Missing city value	Filled with the most frequent city (mode)
Salary outlier (George, KES 900,000)	Reviewed against IQR bounds and corrected to KES 90,000
Inconsistent date format	Parsed as day-first dates and standardized

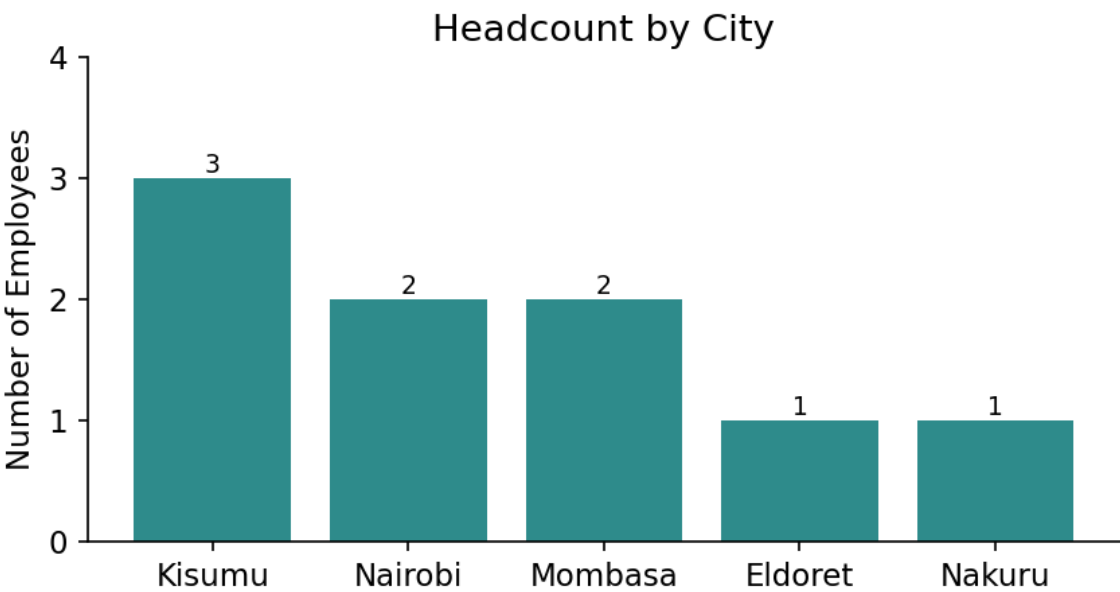
Key Metrics

9 Total Employees	KES 554,500 Total Salary Expense	KES 61,611 Average Salary	30.3 Average Age
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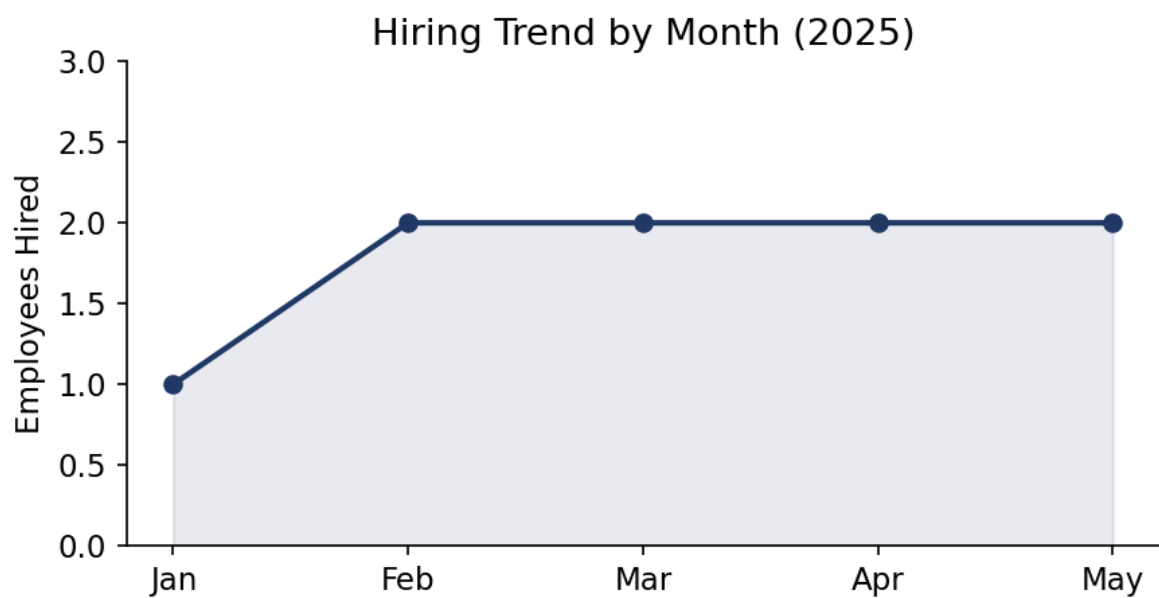
Salary Distribution by City



Headcount by City



Hiring Trend by Month



Employee Salary Ranking

Name	City	Salary (KES)
George	Nairobi	90,000
Diana	Kisumu	70,000
Bob	Mombasa	62,000
Jamal	Kisumu	61,000
Charlie	Kisumu	59,500
Hassan	Mombasa	58,000
Alice	Nairobi	55,000
Faith	Eldoret	51,000
Evan	Nakuru	48,000

Key Insights

- Nairobi has the highest average salary (KES 72,500), driven by senior hires, while Nakuru has the lowest (KES 48,000).
- Kisumu has the largest headcount (3 employees), followed by Nairobi and Mombasa (2 each).
- Hiring was steady through the first five months of 2025, with two new hires in every month except January.

- The total monthly salary expense across all employees is KES 554,500, averaging KES 61,611 per employee.

Prepared as part of a data analytics portfolio project | Python, MySQL, SQL, Power BI